

The consecutive spacing–ratio of nucleosome arrays: high regularity from steric renewal, not level repulsion

Ruqing Chen

GUT Geoservice Inc., Montréal, Québec, Canada

ruqing.chen@gutgeoservice.com

June 25, 2026

Abstract

Genome-wide maps place nucleosome dyads in arrays with a characteristic ~ 150 – 165 bp repeat, and nucleosome spacing, phasing and periodicity have been studied exhaustively. Here we examine these arrays with a statistic borrowed from random-matrix theory (RMT): the unfold-free consecutive spacing ratio $\langle r \rangle$, which discriminates uncorrelated (Poisson, $\langle r \rangle = 0.386$), level-repulsive (Gaussian orthogonal/unitary ensembles, $0.531/0.603$) and rigidly regular ($\langle r \rangle \rightarrow 1$) sequences without spectral unfolding. Applied to base-pair-resolution chemical maps of *Saccharomyces cerevisiae* and *Schizosaccharomyces pombe*, the dyad arrays give $\langle r \rangle \approx 0.78$ and 0.81 — far above every RMT ensemble, superficially suggesting strong order. A permutation test overturns this reading: randomising the order of consecutive linker lengths leaves $\langle r \rangle$ unchanged (within 0.001), and the lag-1 spacing autocorrelation is ≈ 0 . The elevated $\langle r \rangle$ is therefore entirely a property of the marginal linker-length distribution — a steric hard-core renewal process — with no sequential level repulsion or spectral rigidity. The result is robust to confidence thresholds and to nucleosome-free-region trimming, and holds in both species, with $\langle r \rangle$ tracking each species’ shorter or longer repeat length. Superposing independent arrays, or randomly thinning a single array, both drive $\langle r \rangle$ back toward the Poisson value, confirming that the order is local and steric. A high spacing ratio in a genomic (or spatial) point pattern thus need not indicate level repulsion: the permutation test is a necessary control.

1 Introduction

In eukaryotes, genomic DNA is packaged into nucleosomes, each wrapping ~ 147 bp of DNA around a histone octamer, separated by short linkers into arrays with a nucleosome repeat length (NRL) of ~ 150 – 200 bp depending on organism and chromatin context. The advent of base-pair-resolution chemical mapping (Brogaard et al., 2012; Moyle-Heyrman et al., 2013) fixed the centre (dyad) of individual nucleosomes directly, superseding the indirect, lower-resolution micrococcal nuclease assays. These maps established that nucleosomes occupy preferred positions separated by near-integral multiples of the DNA helical repeat, and that much of the genome-wide positioning is explained by steric exclusion: a hard-core “statistical positioning” in which an array is set by a barrier and the finite footprint of the particles (Kornberg & Stryer, 1988; Chereji et al., 2018). Spacing distributions, phasograms, autocorrelation functions and Fourier periodicities of these arrays have since been characterised in great detail.

A complementary and, to our knowledge, previously unused lens on an ordered sequence is the spectral statistics of random-matrix theory (RMT). The spacings between consecutive levels of a sequence carry information about the underlying process: an uncorrelated (or superposed)

sequence follows Poisson statistics, whereas a sequence with *level repulsion* — spacings that avoid small values and are sequentially anti-correlated — follows the Wigner–Dyson statistics of the Gaussian ensembles (Bohigas et al., 1984; Berry & Tabor, 1977; Mehta, 2004). A convenient, unfolding-free measure is the ratio of consecutive spacings (Oganesyan & Huse, 2007; Atas et al., 2013),

$$r_i = \frac{\min(s_i, s_{i+1})}{\max(s_i, s_{i+1})}, \quad s_i = x_{i+1} - x_i, \quad (1)$$

whose mean $\langle r \rangle$ takes the reference values 0.386 (Poisson), 0.531 (Gaussian orthogonal ensemble, GOE), and 0.603 (Gaussian unitary ensemble, GUE), and tends to 1 for a perfectly regular “picket-fence” sequence. Because $\langle r \rangle$ uses only ratios of adjacent gaps it requires no spectral unfolding, making it well suited to inhomogeneous empirical sequences.

A point that is essential below: $\langle r \rangle$ is *two-ended*. The Poisson value 0.386 is the limit of an *uncorrelated* (or superposed) sequence; it is not the value of a single, clean, regular sequence, which instead tends toward 1. A large $\langle r \rangle$ can therefore arise in two very different ways — from genuine sequential level repulsion (as in a chaotic quantum spectrum) or, trivially, from a narrow marginal spacing distribution with no sequential structure at all. Distinguishing the two requires a control that destroys sequential order while preserving the marginal: a permutation (shuffle) test.

Here we compute $\langle r \rangle$ for nucleosome dyad arrays for the first time, using the high-resolution chemical maps of *S. cerevisiae* (Brogaard et al., 2012) and *S. pombe* (Moyle-Heyrman et al., 2013), and ask where the arrays fall and, critically, whether any departure from Poisson reflects sequential repulsion or merely the marginal linker distribution.

2 Data and methods

Data. We use the published unique (distinct-nucleosome) chemical maps: *S. cerevisiae*, 67,548 dyads on 16 chromosomes (Brogaard et al., 2012); *S. pombe*, 75,828 dyads on 3 chromosomes (Moyle-Heyrman et al., 2013). Both list, per nucleosome, a chromosome, a dyad position (bp) and a positioning score. We deliberately use the *unique* maps rather than the accompanying “redundant” maps: the latter enumerate multiple overlapping candidate positions per genomic nucleosome (median centre-to-centre distance ~ 12 bp, well below the 147 bp footprint) and so do not represent distinct particles; they are unsuitable for inter-nucleosome spacing.

Spacing ratio and permutation test. For each chromosome we sort dyad positions, form consecutive spacings s_i , and compute $\langle r \rangle$ from Eq. (1), pooling the r_i within chromosomes (never across chromosome boundaries). The *shuffle test* randomly permutes the order of $\{s_i\}$ within each chromosome and recomputes $\langle r \rangle$. Permutation preserves the marginal spacing distribution exactly while destroying any sequential correlation; it is identical to drawing an i.i.d. renewal process from the empirical linker-length distribution. If $\langle r \rangle_{\text{obs}} \approx \langle r \rangle_{\text{shuf}}$, the value of $\langle r \rangle$ is set entirely by the marginal (scalar) and carries no sequential level repulsion. We also report the lag-1 autocorrelation of the spacings, which is clearly negative for a rigid (repulsive) spectrum and ≈ 0 for a renewal process.

Superposition and thinning. As additional diagnostics we (i) overlay k independent arrays in a shared coordinate and recompute $\langle r \rangle$ as a function of k , and (ii) randomly thin the array (by retaining only the highest-confidence nucleosomes) and recompute $\langle r \rangle$. Both operations are expected to drive a locally-ordered sequence toward the Poisson value.

3 Results

3.1 A high but shuffle-invariant spacing ratio

For *S. cerevisiae* the pooled spacing ratio is $\langle r \rangle = 0.783$, essentially identical across all 16 chromosomes (0.774–0.789; Table 1). This lies far above the Poisson, GOE and GUE reference values, in the quasi-regular regime approaching the picket-fence limit (Fig. 1a). Taken alone, the number would suggest an exceptionally ordered sequence.

The permutation test removes this interpretation. Shuffling the order of the linker lengths gives $\langle r \rangle_{\text{shuf}} = 0.782 \pm 0.0004$, differing from the observed value by +0.0006 — i.e. the two are indistinguishable (Fig. 1a). The lag-1 spacing autocorrelation is +0.05, near zero and, if anything, weakly positive rather than the negative value expected of a rigid spectrum. The high $\langle r \rangle$ is therefore a property of the marginal spacing distribution alone: an i.i.d. renewal process drawn from the same linker distribution reproduces it exactly. There is no sequential level repulsion.

3.2 The spacing distribution

The origin of the high $\langle r \rangle$ is evident in the spacing distribution (Fig. 1b): a sharp mode at 164 bp (the NRL), a hard lower edge near the 147 bp nucleosome footprint (steric exclusion), a fine ~ 10 bp comb from rotational positioning, and a heavy tail of large gaps from nucleosome-free regions (NFRs) and skipped nucleosomes (overall CV = 0.61). The interquartile range is narrow (147–187 bp): most consecutive spacings are both close to the mode, so most r_i are close to 1, which is what drives $\langle r \rangle$ upward. The large-gap tail inflates the coefficient of variation but contributes only a minority of low- r pairs. We note that the ~ 10 bp comb and the integer-bp resolution render the spacings partly discretised rather than drawn from a continuous density, for which the reference values of Eq. (1) are strictly defined. This does not affect the comparison that matters here: $\langle r \rangle_{\text{obs}}$ and $\langle r \rangle_{\text{shuf}}$ are computed on the identically discretised spacings, so the shuffle test — and hence the scalar verdict — is unaffected by discretisation. The permutation test is in this sense robust to non-continuous spacing data.

3.3 Robustness

The scalar (shuffle-invariant) verdict is robust (Table 2). Excluding the large NFR gaps to isolate the bulk array raises $\langle r \rangle$ to 0.81–0.84 (the bulk array is an even tighter steric renewal) while leaving $\langle r \rangle_{\text{obs}} \approx \langle r \rangle_{\text{shuf}}$. Conversely, randomly thinning the array by keeping only the highest-confidence nucleosomes lowers $\langle r \rangle$ monotonically toward the Poisson value ($0.59 \rightarrow 0.48 \rightarrow 0.36$ for the top 50/25/10%), again with $\langle r \rangle_{\text{obs}} \approx \langle r \rangle_{\text{shuf}}$ throughout. Random deletion of points from the array thus destroys the local steric order and returns the sequence to Poisson — the order is local, not a long-range sequential property.

3.4 Superposition drives the ratio to Poisson

Overlaying k independent arrays in a shared coordinate drives $\langle r \rangle$ from 0.78 at $k = 1$ toward the Poisson value as k increases ($0.78 \rightarrow 0.50 \rightarrow 0.43$ for $k = 1, 2, 4$; Fig. 1c), for both an overlay of distinct chromosomes and an overlay of i.i.d. renewals with the empirical marginal. (The plateau slightly above 0.386 reflects the residual hard core, which survives superposition.) Superposition of independent ordered sequences therefore behaves as expected for uncorrelated pooling, confirming that the single-array order is local and additive only at the level of the marginal. This is the well-known convergence of a superposition of independent sequences toward Poisson statistics: the

Table 1: Consecutive spacing ratio $\langle r \rangle$ for nucleosome dyad arrays. Reference values: Poisson 0.386, GOE 0.531, GUE 0.603, regular $\rightarrow 1$.

Species	N pairs	$\langle r \rangle_{\text{obs}}$	$\langle r \rangle_{\text{shuf}}$	lag-1 autocorr.	median spacing (CV)
<i>S. cerevisiae</i>	67,516	0.783	0.782	+0.05	164 bp (0.61)
<i>S. pombe</i>	75,825	0.811	0.811	+0.07	151 bp (0.47)

Table 2: Robustness of the shuffle-invariant verdict (*S. cerevisiae*). $\langle r \rangle_{\text{obs}} \approx \langle r \rangle_{\text{shuf}}$ throughout: the value moves with the marginal, but the scalar (non-repulsive) character does not.

Treatment	N pairs	$\langle r \rangle_{\text{obs}}$	$\langle r \rangle_{\text{shuf}}$
all unique (baseline)	67,516	0.783	0.782
bulk array, exclude gaps > 300 bp	65,204	0.806	0.807
bulk array, exclude gaps > 250 bp	62,276	0.825	0.826
bulk array, exclude gaps > 220 bp	58,900	0.842	0.841
top 50% confidence (thinned)	33,750	0.593	0.581
top 25% confidence (thinned)	16,867	0.480	0.464
top 10% confidence (thinned)	6,729	0.358	0.362

high $\langle r \rangle$ of a single array is a local hard-core property rather than a long-range feature, and does not survive independent mixing.

3.5 Cross-species confirmation

S. pombe gives the same verdict (Fig. 2; Table 1): $\langle r \rangle_{\text{obs}} = 0.811$, uniform across its three chromosomes, with $\langle r \rangle_{\text{shuf}} = 0.811$ (difference -0.0003) and lag-1 autocorrelation $+0.07$. Its spacing distribution is shifted to shorter, tighter values (median 151 bp vs. 164 bp; CV 0.47 vs. 0.61), consistent with the species’ known shorter NRL and ~ 4 –5 bp linker preference. The *value* of $\langle r \rangle$ is correspondingly *higher* in *S. pombe* (a tighter marginal yields a larger ratio), even as the scalar, non-repulsive character is unchanged. That $\langle r \rangle$ shifts with the marginal exactly as the shuffle interpretation predicts — while the verdict remains species-invariant — is itself a strong internal consistency check.

4 Discussion

The consecutive spacing ratio of nucleosome dyad arrays is high ($\langle r \rangle \approx 0.78$ –0.81, above even the GUE value) yet entirely *scalar*: it is reproduced to three decimal places by an i.i.d. renewal with the same linker-length distribution, and the spacings carry no negative sequential correlation. The arrays are, in this precise statistical sense, a one-dimensional steric renewal (hard-rod) process — the quantitative signature of the statistical-positioning and steric-occlusion pictures of nucleosome organisation (Kornberg & Stryer, 1988; Chereji et al., 2018), now expressed in the language of level-spacing statistics. In statistical-mechanical terms the array behaves, to first order, as a one-dimensional Tonks gas: a sequence of mutually impenetrable hard rods of the ~ 147 bp footprint placed along the genome (Tonks, 1936). The analogy is only first order, however. An equilibrium Tonks gas has an exponential distribution of gaps above the rod length, whereas the observed linker distribution is sharply peaked at a preferred repeat (Fig. 1b), indicating an additional, remodeler-

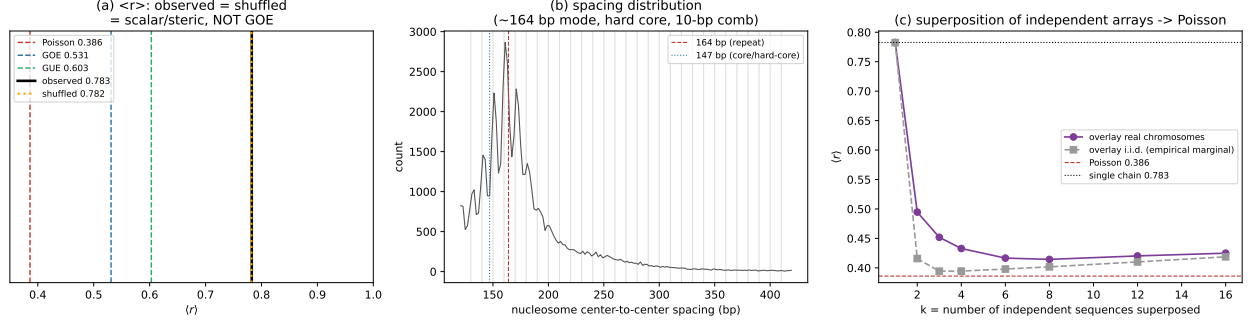


Figure 1: Nucleosome dyad arrays in *S. cerevisiae*. **(a)** The observed spacing ratio (0.783) coincides with the shuffled value (0.782) and sits far above the Poisson, GOE and GUE references: the regularity is scalar, not level repulsion. **(b)** Spacing distribution: a 164 bp mode, a hard edge near the 147 bp footprint, a ~ 10 bp rotational comb and a large-gap (NFR) tail. **(c)** Overlaying k independent arrays drives $\langle r \rangle$ from 0.78 toward the Poisson value (real-chromosome overlay and i.i.d. overlay agree).

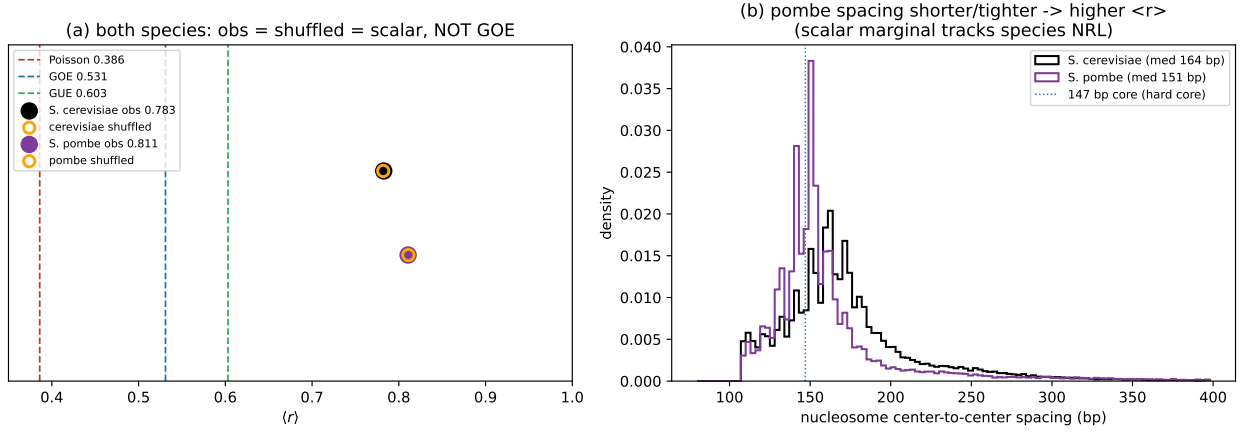


Figure 2: Cross-species comparison. **(a)** In both *S. cerevisiae* and *S. pombe* the observed $\langle r \rangle$ coincides with the shuffled value (orange rings) and lies far above the RMT ensembles — scalar, not repulsive — in both organisms. **(b)** The *S. pombe* spacing distribution is shorter and tighter (median 151 bp) than *S. cerevisiae* (164 bp); the tighter marginal yields the higher $\langle r \rangle$, exactly as the shuffle interpretation predicts.

set preferred spacing superimposed on the pure steric exclusion. That extra spacing regularity is itself scalar — it is present, unchanged, in the shuffled sequence — and so likewise produces no level repulsion. A genuinely level-repulsive (Wigner–Dyson) array would instead show $\langle r \rangle_{\text{obs}}$ *differing* from $\langle r \rangle_{\text{shuf}}$ and a clearly negative lag-1 autocorrelation (spectral rigidity); neither is observed.

The main methodological lesson is cautionary. Because $\langle r \rangle$ is two-ended, a value far above the Poisson reference — even above the GUE ensemble — does not by itself indicate order of the kind RMT describes. Here essentially all of the elevation is marginal: it follows from a narrow, hard-core spacing distribution and would be produced by shuffled (sequence-free) data. The permutation test is therefore not an optional refinement but a necessary control whenever $\langle r \rangle$ is applied to genomic or spatial point patterns, where narrow marginal distributions are common. The two auxiliary diagnostics make the same point from the opposite direction: superposing independent arrays, and randomly thinning a single array, both return $\langle r \rangle$ toward Poisson, as expected when the order is local and steric rather than a long-range sequential property.

Finally, the cross-species comparison sharpens the interpretation rather than merely repeating

it. *S. pombe*, with its shorter repeat length and $\sim 4\text{--}5$ bp linker preference, has a tighter marginal and correspondingly a *higher* $\langle r \rangle$, while the scalar, non-repulsive verdict is unchanged. The ratio thus behaves as a faithful readout of the marginal spacing distribution across two highly diverged genomes — useful as a compact, unfolding-free summary of array tightness, provided it is not over-read as level repulsion.

We emphasise what is and is not new. The regularity, periodicity and steric basis of nucleosome spacing are well established; we claim no discovery there. The contribution is the level-spacing-ratio (RMT) characterisation of these arrays and the explicit demonstration — via permutation, superposition and thinning — that their high $\langle r \rangle$ is scalar rather than repulsive. The same toolkit, with the same controls, applies directly to other one-dimensional genomic feature sequences.

Data availability

S. cerevisiae dyad positions: Brogaard et al. (2012), GEO accession GSE36063 / Nature Supplementary Table 2 (unique map). *S. pombe* dyad positions: Moyle-Heyrman et al. (2013), PNAS Supporting Information Dataset S1 (unique map).

Code availability. Analysis code, derived spacing data and figures are openly available at <https://github.com/Ruqing1963/nucleosome-spacing-ratio> and archived on Zenodo (DOI assigned on release).

Acknowledgements

The author thanks the laboratories of J. Widom and colleagues for making the base-pair-resolution chemical maps publicly available.

References

- Y. Y. Atas, E. Bogomolny, O. Giraud, and G. Roux. Distribution of the ratio of consecutive level spacings in random matrix ensembles. *Phys. Rev. Lett.* **110**, 084101 (2013).
- M. V. Berry and M. Tabor. Level clustering in the regular spectrum. *Proc. R. Soc. Lond. A* **356**, 375–394 (1977).
- O. Bohigas, M. J. Giannoni, and C. Schmit. Characterization of chaotic quantum spectra and universality of level fluctuation laws. *Phys. Rev. Lett.* **52**, 1–4 (1984).
- K. Brogaard, L. Xi, J.-P. Wang, and J. Widom. A map of nucleosome positions in yeast at base-pair resolution. *Nature* **486**, 496–501 (2012).
- R. V. Chereji, S. Ramachandran, T. D. Bryson, and S. Henikoff. Precise genome-wide mapping of single nucleosomes and linkers in vivo. *Genome Biol.* **19**, 19 (2018).
- R. D. Kornberg and L. Stryer. Statistical distributions of nucleosomes: nonrandom locations by a stochastic mechanism. *Nucleic Acids Res.* **16**, 6677–6690 (1988).
- M. L. Mehta. *Random Matrices*, 3rd ed. Elsevier/Academic Press, Amsterdam (2004).

- G. Moyle–Heyrman, T. Zaichuk, L. Xi, Q. Zhang, O. C. Uhlenbeck, R. Holmgren, J. Widom, and J.-P. Wang. Chemical map of *Schizosaccharomyces pombe* reveals species–specific features in nucleosome positioning. *Proc. Natl. Acad. Sci. USA* **110**, 20158–20163 (2013).
- V. Oganessian and D. A. Huse. Localization of interacting fermions at high temperature. *Phys. Rev. B* **75**, 155111 (2007).
- L. Tonks. The complete equation of state of one, two and three–dimensional gases of hard elastic spheres. *Phys. Rev.* **50**, 955–963 (1936).